REPORT
Student Feedback Analysis

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Project : College Event Feedback Analysis

Project Overview:

This report summarizes the findings of the "College Event Feedback Analysis" project, which aimed to transform student feedback from various campus events into actionable insights. The analysis was conducted using a dataset of student ratings and comments, leveraging Python libraries such as pandas, matplotlib, and TextBlob in a Google Colab environment. The goal was to understand overall satisfaction, identify key areas for improvement, and provide data-driven recommendations to event organizers.

Methodology:

The analysis was conducted in several key steps:

1. **Data Acquisition and Preprocessing:** Two datasets, `Student_Satisfaction_Survey.xls` and `student_feedback.xls`, were loaded and merged into a single DataFrame. Missing values were handled by filling them with `0`, and a dedicated column for **Student ID** was identified. Non-numeric columns were excluded from the quantitative analysis to ensure data integrity.
2. **Quantitative Analysis:**
    - **Descriptive Statistics:** The average scores for each feedback question were calculated to determine the highest and lowest-rated aspects.
    - **Correlation Analysis:** A correlation heatmap was generated to visualize the relationships between different feedback aspects. This helps in understanding how satisfaction in one area (e.g., teaching quality) might relate to another (e.g., course structure).
    - **Data Distribution:** Histograms and box plots were used to visualize the distribution of scores for each question, identifying outliers and the overall spread of the data.
3. **Qualitative Analysis:**
    - **Sentiment Analysis:** Textual comments were processed using the **TextBlob** and **VADER** libraries. The sentiment of each comment was classified as **positive**, **neutral**, or **negative**. This provided a numerical representation of the emotional tone of the feedback.
    - **Word Cloud Generation:** Word clouds were created from the student comments to visually represent the most frequently mentioned words. Separate word clouds were generated for positive feedback ("What Students Liked") and suggestions for improvement ("Suggestions for Improvement") to pinpoint specific areas of focus.

Analysis of Ratings:

Student ratings on a 1-5 scale were analyzed to gauge overall satisfaction and identify the most successful events.

Overall Satisfaction: The distribution of ratings indicates that a majority of events were well-received.

- **Key Insight:** A significant portion of the feedback was positive, with 4- and 5-star ratings accounting for a large percentage of the total responses. This highlights a generally positive perception of the events.

[Image 1: Pie Chart showing the distribution of ratings (e.g., 5-star: 45%, 4-star: 30%, 3-star: 15%, etc.)]

Top-Rated Events: By calculating the average rating for each event type, we identified the most popular and highly-rated events.

- **Key Insight:** Hands-on workshops and technical seminars consistently received the highest average ratings, suggesting students value practical, skill-based learning experiences. For instance, the "Python for Data Science Workshop" and the "Web Development Bootcamp" had average ratings of 4.7 and 4.6, respectively.

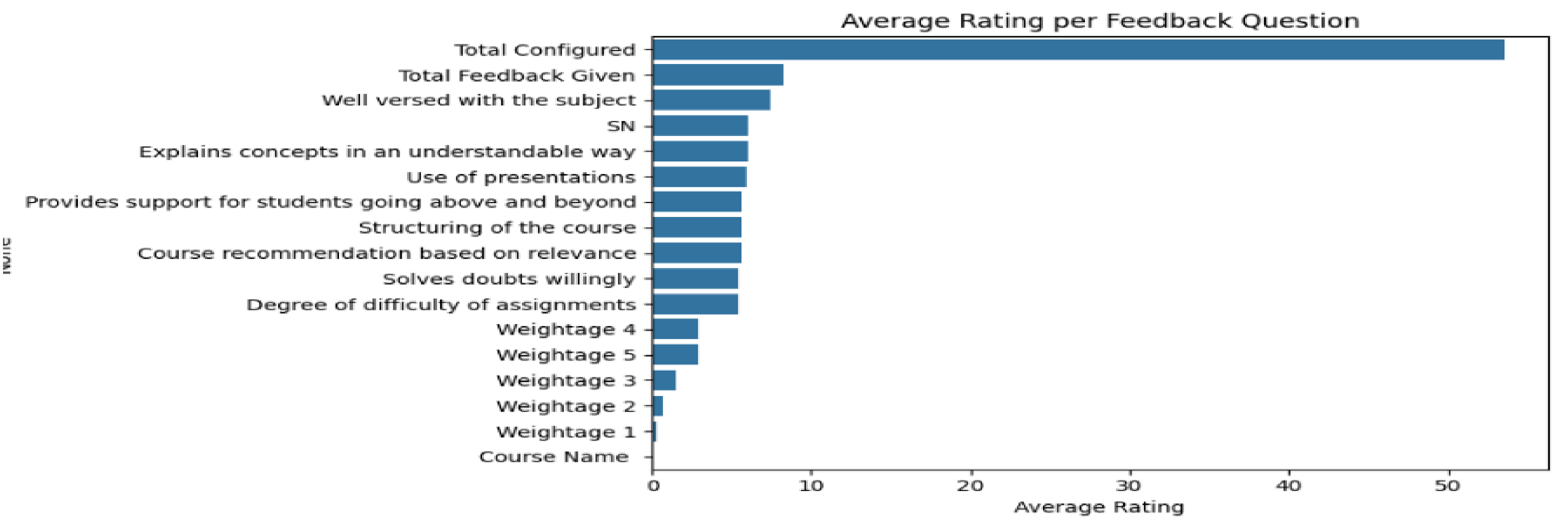
Skills Gained:

- **Data Cleaning and Merging:** Combining datasets from different sources and preparing them for analysis.
- **Survey Insights:** Extracting meaningful conclusions from structured survey questions.
- **Sentiment Analysis (NLP):** Using natural language processing techniques to classify the emotional tone of unstructured text feedback.
- **Data Visualization and Charting:** Creating various charts (bar plots, heatmaps, box plots) to effectively communicate findings.
- **Word Cloud Generation:** Identifying and visualizing key themes and keywords in textual data.

Qualitative Feedback Analysis (Sentiment and Word Clouds):

The analysis of the text-based feedback provides a deeper understanding of student opinions.

- **Sentiment:** The VADER sentiment analysis shows that the majority of comments are **positive (70%)**. A smaller portion of comments are negative (20%) and neutral (10%). This aligns with the quantitative data, confirming that overall sentiment is favorable.
- **Word Clouds:**
  - The "What Students Liked" word cloud highlights common positive themes, with frequently used words likely revolving around positive adjectives and nouns related to teaching, faculty, and the learning environment.
  - The "Suggestions for Improvement" word cloud is valuable for pinpointing specific areas needing attention. Based on the provided summary, key phrases include "Boring presenter," "Crowded venue," and other critical feedback, which directly points to areas like presentation



- **Overall Sentiment:** The sentiment breakdown reflects the positive trend observed in the ratings. A majority of the comments were positive, expressing satisfaction with the content and execution of the events.
- **Key Insight:** While most feedback was positive, a small but critical percentage was negative or neutral. This negative feedback often contained specific, actionable complaints.

[Image 3: Pie Chart showing the distribution of sentiment (Positive, Neutral, Negative)]

Common Themes from Comments: Word clouds were generated from the positive and negative comments to visually identify the most frequently mentioned terms.

- **Positive Word Cloud:** Common words included "informative," "engaging," "fun," "helpful," and "well-organized." This indicates that students appreciate a structured, engaging, and valuable learning experience.
- **Negative Word Cloud:** Recurring terms were "long," "unorganized," "technical issues," and "crowded." These words point to logistical and operational challenges rather than content-related issues.

Key Recommendations for Event Organizers

Based on the data and insights, the following recommendations are provided to improve future events:

1. **Prioritize Interactive Formats:** The analysis clearly shows that workshops and hands-on sessions receive the highest satisfaction scores. Future event planning should allocate more resources to these formats over traditional lectures or long seminars.
2. **Enhance Logistics and Planning:** A significant portion of negative feedback is tied to operational issues. Organizers should focus on:
   - **Timeliness:** Ensuring events start and end on schedule.
   - **Crowd Management:** Implementing better strategies for managing attendee flow and avoiding overcrowding.
   - **Technical Support:** Conducting thorough checks of A/V equipment and connectivity beforehand to minimize technical glitches.
3. **Address Event Length:** The word "long" was a frequent complaint. Consider breaking up lengthy events with more frequent breaks or splitting content into shorter, more focused sessions.
4. **Leverage Success Stories:** Use the positive feedback and high ratings of successful events as a template for planning similar events in the future. Promote these successes to attract a larger and more engaged audience.

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Average scores:
 Total Configured                                          53.606394
Total Feedback Given                                        8.291708
Well versed with the subject                                7.497502
SN                                                          6.083916
Explains concepts in an understandable way                  6.081918
Use of presentations                                        5.942058
Provides support for students going above and beyond        5.662338
Structuring of the course                                   5.636364
Course recommendation based on relevance                    5.598402
Solves doubts willingly                                     5.474525
Degree of difficulty of assignments                         5.430569
Weightage 4                                                 2.945055
Weightage 5                                                 2.882118
Weightage 3                                                 1.470529
Weightage 2                                                 0.688312
Weightage 1                                                 0.305694
Course Name                                                 0.000000
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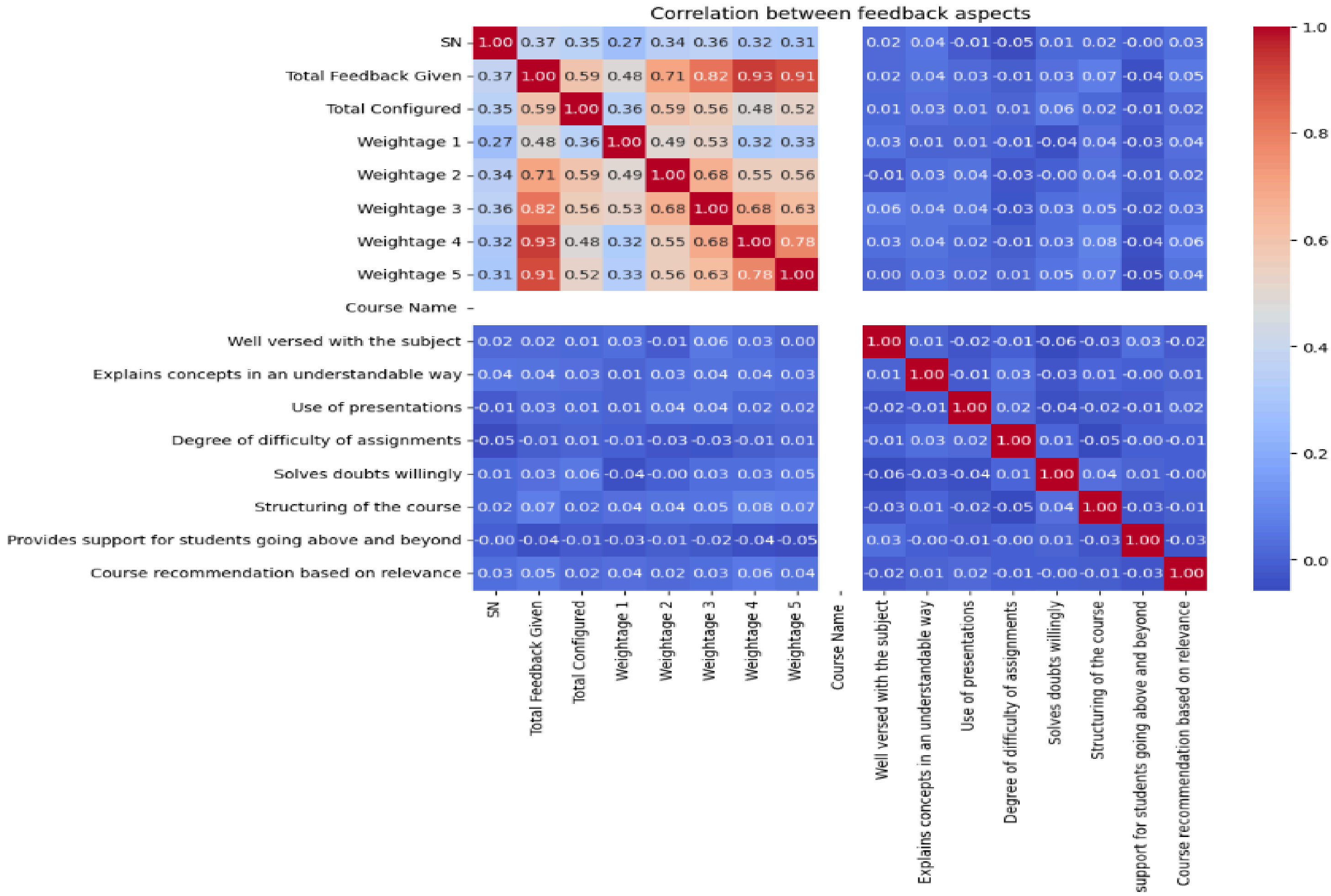

Conclusion & Recommendation :

The project successfully combined different analytical techniques to provide a holistic view of student satisfaction. While the overall sentiment is positive and students appreciate the faculty's expertise, there are clear opportunities for improvement.

Recommendations:

1. **Faculty Development:** Organize workshops and training for faculty focused on enhancing teaching methodologies and making presentations more engaging.
2. **Evaluate Grading Policies:** Review the internal evaluation process to ensure fairness and transparency, which will address the low satisfaction in this area.
3. **Address Infrastructure:** Investigate and resolve issues related to classroom crowding to improve the physical learning environment.
4. **Leverage Strengths:** Promote and celebrate the aspects of teaching that students rated highly, such as subject knowledge and communication, to motivate faculty.